

Who Moves and Who Benefits? The Value of Migration Across Households

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Abstract

This paper studies how the value of migration opportunities and migration rates differ across heterogeneous households. To do so, I develop a quantitative life-cycle spatial equilibrium model with multiple locations, migration costs, wealth accumulation, and housing tenure choice. Using the model, I measure the value of migration as the additional consumption households would require to maintain the same level of welfare in the absence of migration opportunities. The average value of migration is estimated to be 2.56% of consumption. However, this average masks substantial heterogeneity across household groups. Both migration values and migration rates are higher for younger households. In contrast, across wealth and tenure groups, the two measures move in different directions. Wealthier households and homeowners exhibit lower migration rates but obtain larger migration values. These differences arise because migration motives and the scope for adjusting consumption and housing through migration vary systematically with age, wealth, and tenure status. I also examine how migration values and migration rates change when regional productivity dispersion increases. While the overall pattern of migration values and migration rates across household groups remains broadly unchanged, average migration values increase and migration rates decline. However, these additional migration gains are not distributed evenly across households and are particularly large for older and wealthier households. These results show that migration rates and migration values do not necessarily move in the same direction. They also show that migration values and the gains arising from changes in the economic environment differ systematically across migration motives and household characteristics.

Keywords: Internal Migration, Migration Value, Spatial Equilibrium, Household Heterogeneity, Housing Choice.

JEL Classification: J6, R10, R13, R31

1 Introduction

Millions of households in the United States relocate each year. Some move to pursue better job opportunities, while others relocate to obtain better housing, live closer to family members, or provide better educational opportunities for their children. These moves may involve short-distance relocations within local areas, but they can also take the form of long-distance migration across states and regions.

Although migration may appear to be a small decision made by individual households, its economic effects extend throughout the broader economy. One simple example is that households' relocation across regions changes local labor supply and housing demand, thereby affecting wages and housing prices. These changes in local prices alter the economic environment faced by households in each region and, in turn, affect household welfare. Therefore, studying internal migration is important for understanding how key economic conditions, including local labor markets, housing markets, and household welfare, evolve over time.

Given its importance, a growing literature has studied migration and location choice from different perspectives. One group of papers focuses on the aggregate effects of migration. For example, [Moretti \(2010\)](#), [Hsieh and Moretti \(2019\)](#), and [Bilal and Rossi-Hansberg \(2021\)](#) study how migration and location choices affect local labor markets, housing markets, and welfare. Another group of papers focuses on the specific role of internal migration. For example, [Yagan \(2014\)](#), [Yagan \(2019\)](#), and [Giannone et al. \(2025\)](#) analyze the insurance role of migration against idiosyncratic or local labor market shocks and its implications for household welfare and policy outcomes.

Despite the broad literature on migration from these different perspectives, much less is known about migration itself: why households move and how migration motives and migration gains differ across households.¹ Understanding the broader consequences of migration is important, but doing so requires understanding migration itself. Without understanding why households choose to migrate and under what conditions migration occurs, it is difficult to fully interpret the economic effects of migration. For example, households that migrate in response to labor market change may have very different motives and economic implications

¹To the best of my knowledge, [Oswald \(2019\)](#) is the only paper that examines the value of migration itself.

from households that migrate to improve their housing conditions.

To better understand migration, it is important to recognize that migration is inherently a heterogeneous decision. Migration incentives can differ across households because household characteristics, such as current wealth and age, are heterogeneous. Therefore, examining migration itself requires looking beyond aggregate migration flows or average migration values. For this reason, I focus on heterogeneous household groups and study how they make migration decisions, which groups move more frequently, and how much value different groups obtain from migration.

In this paper, I use a model to examine the migration motives of heterogeneous households and the migration values generated by these motives. One of the main contributions of this paper is to develop and solve a quantitative life-cycle spatial equilibrium model that combines idiosyncratic productivity risk and wealth accumulation within the quantitative spatial framework of [Redding and Rossi-Hansberg \(2017\)](#) and [Greaney \(2026\)](#).² Locations differ in productivity, amenities, labor market risk, and housing market conditions. The economy consists of overlapping generations of risk-averse households that face idiosyncratic income shocks and mortality risk during their working lives. In each period, households make forward-looking decisions regarding non-housing consumption, housing consumption, saving, location choice, tenure status. Markets are incomplete, households cannot fully insure against idiosyncratic productivity (income) risk, which generates a precautionary saving motive. Households accumulate wealth through liquid assets and owner-occupied housing and make saving and borrowing decisions subject to a borrowing constraint. Migration is subject to migration costs which is utility costs. On the production side, each location produces a tradable good using local labor and a decreasing-returns technology. Local productivity is determined endogenously through agglomeration forces.

The model features a large state space because households differ in wealth, idiosyncratic labor productivity, location, age, and tenure status. To address this challenge, I adopt the mixed time structure developed by [Greaney \(2026\)](#). Continuous saving decisions are solved using the continuous-time dynamic programming approach of [Achdou et al. \(2022\)](#), which

²[Redding and Rossi-Hansberg \(2017\)](#) provides the standard framework for quantitative spatial models, and I directly adopt the multi-location structure with discrete choices from [Greaney \(2026\)](#).

substantially reduces computation time. In contrast, discrete choices such as idiosyncratic productivity (income) shock realizations, location decisions, and housing adjustments are modeled in discrete time. This approach efficiently allocates computational burden across different decisions. The mixed time structure greatly reduces the number of discrete choice problems that must be solved in the life-cycle model and significantly improves computational efficiency.

The model is calibrated using data from 50 metropolitan areas in the US and provides a good overall fit to the data. Despite assuming that migration costs are identical across households, the model successfully replicates the heterogeneous migration patterns observed across age and tenure groups. In addition, the model matches several untargeted moments reasonably well, including the wealth distribution, the home value-to-wealth ratio, homeownership rates, and migration patterns across locations.

I use the model to quantify the value of the migration option and the migration rate. The average value of the migration option is 2.56% of consumption. This result implies that, in the absence of migration opportunities, households would require an additional 2.56% increase in consumption to attain the same level of welfare. Equivalently, migration opportunities generate welfare gains equal to 2.56% of consumption. The model also matches the aggregate migration rate observed in the data, which is 2.55%.

However, these aggregate statistics mask substantial heterogeneity across household groups. Examining migration values and migration rates across heterogeneous households reveals several important patterns. First, both migration values and migration rates decline with age. Young households exhibit both the highest migration values and the highest migration rates relative to older households. Second, the relationship differs sharply across wealth groups. Migration values increase with wealth, whereas migration rates decrease with wealth. A similar pattern emerges across tenure groups. Among younger households, renters tend to have similar migration values than homeowners. However, as households age, renters continue to exhibit higher migration rates, while homeowners often obtain migration values that are comparable to, or even larger than, those of renters.

These findings provide two important implications. The first implication is the finding itself: migration rates and the welfare value of migration opportunities do not necessarily

move together. Migration rates reflect households' revealed responses to differences in local wages, housing prices, and migration costs. Migration values, in contrast, depend on how much households can improve their consumption and housing expenditure by exercising the migration option, given their current conditions, such as wealth and age. As a result, the mechanisms that determine migration rates and migration values need not be the same. Therefore, observed migration frequencies alone do not provide sufficient information about which households obtain the largest welfare gains from migration opportunities.

The second implication is that housing market conditions are a key determinant of the value of migration opportunities. The value of migration does not arise solely from access to higher wages. It also reflects the possibility of relocating to lower-cost housing markets and adjusting housing consumption. Young low-wealth households are primarily motivated by investment considerations and tend to move toward high-wage locations in order to improve future economic outcomes. In contrast, for older and wealthier households, housing adjustment becomes more important. These households place greater value on moving to locations with lower housing costs, where they can improve current living conditions. In particular, the migration gains associated with housing size adjustment become increasingly important at higher wealth levels.

I then use the model to examine how migration values and migration rates change when regional disparities become larger. Specifically, I consider an environment in which productivity dispersion increases, generating larger differences in wages and housing prices across locations.³ In such an environment, productive locations become more attractive while less productive locations become relatively less attractive. Since migration behavior depends on local economic conditions, understanding how migration values and migration rates respond to these changes provides important insights into the welfare consequences of spatial inequality.

The results show that larger regional disparities increase the welfare value of migration opportunities because households face greater potential gains from relocating across locations. At the same time, migration rates decline because larger housing price gaps between pro-

³Wages are directly influenced by productivity, while housing prices are indirectly influenced by migration responses to changes in the productivity distribution.

ductive and less productive cities make relocation more difficult. However, the additional migration value generated by the new environment is not distributed evenly across households.⁴

The additional gain from the environmental change in migration value becomes larger with age and is particularly pronounced among old and high-wealth households. Since tenure status is closely related to wealth, older homeowners also receive larger additional gains than renters. These findings suggest that although greater regional disparities increase the value of migration opportunities overall, the benefits are distributed unevenly across households according to age, wealth, and housing status.

Related Literature There is a growing body of literature that introduces individuals' location choice and its of spatial heterogeneity. This topic is summarized by [Moretti \(2010\)](#), [Halket \(2012\)](#), [Halket and Vasudev \(2014\)](#), [Ganong and Shoag \(2017\)](#), [Redding and Rossi-Hansberg \(2017\)](#), [Walsh \(2019\)](#), [Hsieh and Moretti \(2019\)](#), [Bilal and Rossi-Hansberg \(2021\)](#), [Crews \(2023\)](#), [Bilal \(2023\)](#), [Lagakos et al. \(2023\)](#), [Crews \(2023\)](#), [Giannone et al. \(2025\)](#), [Greaney et al. \(2025\)](#), and [Greaney \(2026\)](#). These papers explore how households choose where to live and the economic implications of these location choices. In particular, [Giannone et al. \(2025\)](#), [Greaney et al. \(2025\)](#), and [Greaney \(2026\)](#) directly model how households choose locations in the presence of rich household heterogeneity, such as wealth, individual productivity, age, and tenure status. These papers also compute how migration rates differ across wealth, age, and tenure groups. Building on this literature, I focus on how migration motives and migration gains differ across heterogeneous households. In particular, I analyze why households with different characteristics choose to move and how much they gain from having the migration option. This framework allows me to distinguish between migration rates and migration values.

The second group of papers focuses more directly on migration decisions and the value of migration opportunities. [Oswald \(2019\)](#) provides a structural interpretation of the value of migration. He finds that, in the absence of migration, a 19% increase in per-period consump-

⁴The additional migration value refers to the difference between migration values in the counterfactual and baseline economies. The overall patterns remain unchanged in both environments: younger households and wealthier households continue to exhibit higher migration values than older and lower-wealth households.

tion would be required to make households indifferent, highlighting the substantial welfare gains generated by migration opportunities against income and housing price shocks. My paper is closely related to this work because I also quantify the value of the migration option across heterogeneous households. While [Oswald \(2019\)](#) focuses on heterogeneity arising from age and homeownership itself, I additionally incorporate wealth accumulation and housing size adjustment decisions. As a result, I show that homeowners can obtain migration values comparable to, or even larger than, those of renters despite having substantially lower migration rates.

Another closely related paper is [Bilal and Rossi-Hansberg \(2021\)](#), which studies migration in a framework where locations differ in their current and future values. The paper shows that low-wealth households tend to choose locations with lower current costs but lower future values, while high-wealth households are more likely to locate in areas with higher current costs and greater future returns. Building on this idea, I introduce age and housing adjustment decisions into the analysis. I find that migration motives differ substantially across wealth and age groups. Young low-wealth households often move toward expensive high-wage locations because of an investment motive, whereas high-wealth households can obtain large utility gains through housing adjustment by moving to locations with lower housing costs. These results highlight that migration patterns depend not only on wealth but also on the interaction between wealth, age, and housing decisions.

The rest of the paper is organized as follows. Section 2 presents empirical evidence on migration rates and migration motives across household groups. Section 3 develops the quantitative spatial equilibrium model. Section 4 describes the calibration strategy and evaluates the model fit. Section 5 quantifies the value of migration and analyzes how migration motives and migration gains differ across wealth, age, and tenure groups. Finally, Section 6 concludes.

2 Empirical Evidence

In this section, I examine how household migration rates vary over time according to individual characteristics and migration motives. Specifically, I show how migration rates differ

across wealth, age, and tenure groups, and investigate which motives, such as job-related or housing-related reasons, are more important for each group.

Although these findings do not directly contribute to uncovering new facts, they show that migration rates vary across heterogeneous household groups and that households move for different reasons. These patterns suggest that migration motives may differ systematically across household groups. In this sense, the empirical evidence indicates that heterogeneous households may gain different values from migration depending on the motives behind their moves.

2.1 Data Description

American Community Survey. To compute migration rates by household group, I primarily use the American Community Survey (ACS) from 2005 to 2019.⁵ The ACS is a repeated cross-sectional survey that collects annual information on individual and household characteristics, including income, home value, age, tenure status, current location, and location one year prior.

Although the ACS does not directly report household wealth, [Di \(2001\)](#) and [Wolff \(2007\)](#) show that home value accounts for the largest component of household wealth. Therefore, I use home value as the first proxy for household wealth. As an alternative proxy, I use home value plus investment income, which is more closely aligned with the definition of wealth in [Section 3](#). To maintain consistency with the model, I restrict the sample to individuals aged 21 to 90. I also define migration at the Metropolitan Statistical Area (MSA) level. A household is classified as a migrant if its current MSA differs from its MSA one year ago. The migration rate is computed as the number of migrants divided by the population in each MSA. All calculations use person weights, and individuals living in group quarters are excluded from the sample.

Current Population Survey. To examine the distribution of migration motives, I use the Current Population Survey (CPS) from 2005 to 2019. I use the same sample period as in

⁵I start from 2005 because MSA definitions revised in 2013 are available in the ACS beginning in 2005. I restrict the sample period to 2019 to exclude the unusual migration patterns during the COVID-19 period.

the ACS to ensure consistency across datasets. The CPS is a repeated cross-sectional survey that collects monthly information on household income, current location, employment status, migration status, and reasons for moving.

I classify the CPS migration-reason variable into four broad categories: job-related, housing-related, family-related, and other reasons. Job-related reasons include moves associated with a new job, job transfer, job search, job loss, commuting, retirement, or other employment-related considerations. Housing-related reasons include moves motivated by homeownership, better housing, neighborhood quality, lower housing costs, or other housing-related considerations. Family-related reasons include changes in marital status, forming a new household, or other family-related considerations. All remaining responses are classified as other reasons.

I then compute the share of each motive among individuals who moved within the previous year. As in the ACS analysis, I restrict the sample to individuals aged 21 to 90 and use person weights in all calculations.

2.2 Migration Patterns by Household Groups

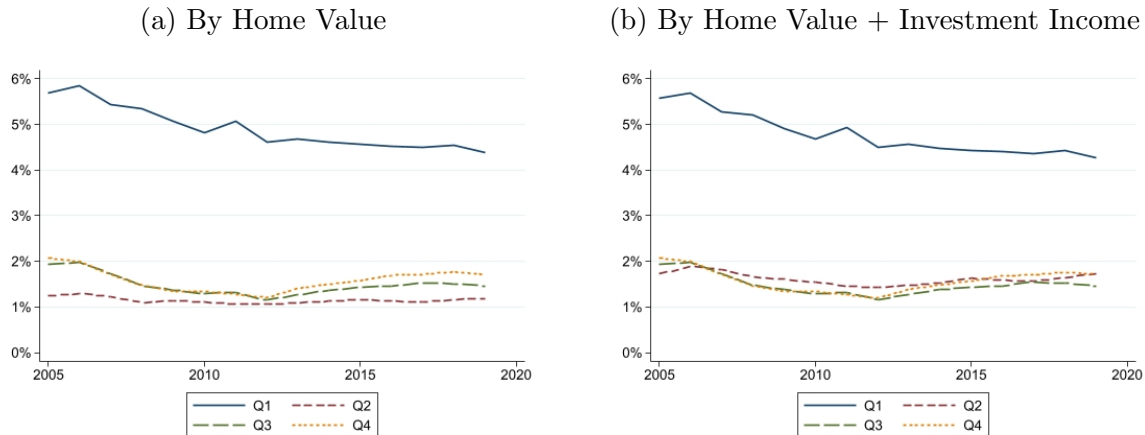
I now report how migration rates vary across different household groups. As discussed above, I classify households by wealth, age, and tenure status.

By Wealth Figure 1 plots migration rates across different wealth groups. I first divide households into four quartiles based on each wealth proxy. Q1 denotes households with the lowest wealth level, while Q4 denotes households with the highest wealth level. Figure 1a reports migration rates across wealth quartiles using home value as the wealth proxy. The figure shows that Q1 households have the highest migration rate throughout the sample period. Although the migration rate of Q1 households gradually declines over time, it remains above 4% in 2019, while the migration rates of the other wealth groups are below 2%.

I then compute migration rates using an alternative wealth proxy, defined as home value plus investment income, where investment income is used as a proxy for financial asset returns. Figure 1b shows a similar pattern. Throughout the sample period, Q1 households have substantially higher migration rates than the other wealth groups. Although their migration rate

also declines over time, it remains higher than the migration rates of all other groups in 2019.

Figure 1: Migration Rates by Wealth Group



Notes: These figures show migration rates by wealth group from 2005 to 2019. In Panel (a), wealth is proxied by home value. In Panel (b), wealth is proxied by the sum of home value and investment income. The data source is the 2005–2019 American Community Survey.

By Demographics Figure 2 displays migration rates across age groups and tenure status. Figure 2a shows differences in migration rates by age. The pattern is clear: younger households are more likely to migrate. Migration rates decline monotonically with age, indicating that older households are less likely to move to another MSA.

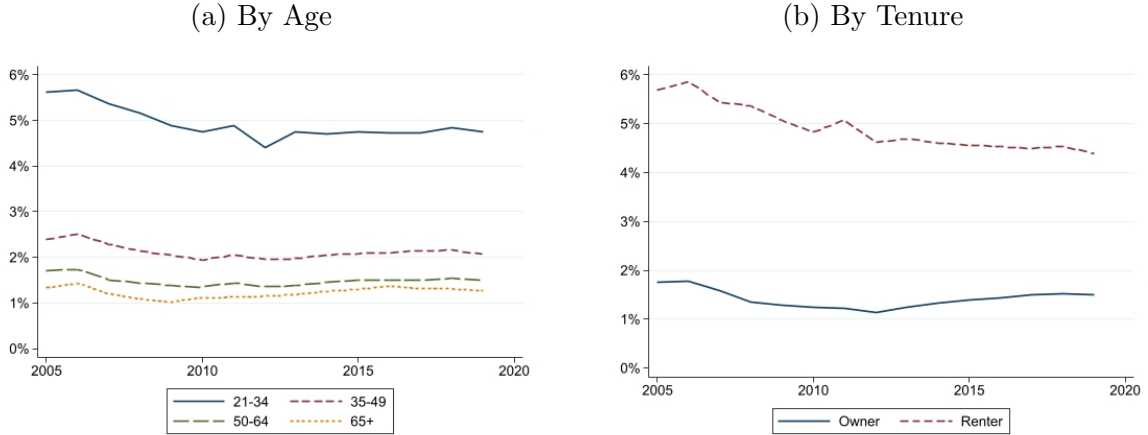
Figure 2b shows the relationship between household tenure status and migration rates. Renters are more likely to migrate than homeowners throughout the sample period. As in the previous figures, migration rates gradually decline over time for both groups.

These patterns are consistent with Molloy et al. (2014), who document the decline in U.S. internal migration both at the aggregate level and across demographic groups. This consistency helps validate the empirical patterns in the data and supports the motivation of this paper.

2.3 Why Households Move?

In this section, I report the reasons why households migrate. As discussed above, I classify migration motives into four groups: job-related, housing-related, family-related, and other

Figure 2: Migration Rate by Demographics



Notes: These figures show migration rates by age group and tenure status from 2005 to 2019. Age groups are constructed using individuals aged 20 to 90. Tenure status is defined by homeownership and renting status. The data source is the 2005–2019 American Community Survey.

reasons. Figure 3 shows migration motives separately for moves across counties within the same state and moves across states. I use these two types of migration instead of MSA-level migration because the current MSA is not identified for most observations in the CPS.

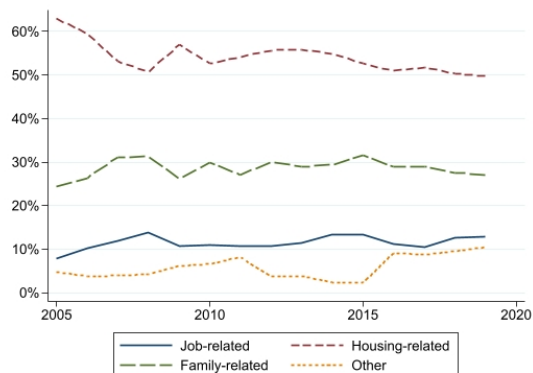
Figure 3a displays the motives of households who move across counties within the same state. The most common motive is housing-related reasons, followed by family-related reasons. Job-related reasons and other reasons account for smaller and relatively similar shares. Figure 3b shows the motives of households who move across states. In contrast to within-state migration, the most common motive for cross-state migration is job-related reasons. The share of job-related moves increases rapidly in the early 2000s and remains close to 50 sample period. The second most common motive is family-related reasons, although housing-related reasons account for a larger share between 2012 and 2016. Other reasons again account for the smallest share.

These patterns suggest that households tend to move for housing-related reasons when they make relatively short-distance moves, while job-related reasons become more important for longer-distance moves. Since this paper studies migration across MSAs, short-distance migration is not necessarily the main object of interest. Nevertheless, both figures show that labor market and housing market considerations are important motives for migration. This empirical pattern supports the motivation of this paper, which emphasizes the role of labor

and housing market differences in shaping migration behavior.

Figure 3: Motives for Migration

(a) Between-County, Within-State Migration



(b) Between-State Migration



Notes: These figures show the motives for migration between 2005 and 2019 among households who moved between counties within the same state and households who moved across states. Panel (a) shows motives for between-county, within-state migration, and Panel (b) shows motives for between-state migration. Job-related reasons include new job or job transfer, looking for work or job loss, easier commute, retirement, and other job-related reasons. Housing-related reasons include wanting to own rather than rent, wanting new or better housing, wanting a better neighborhood, cheaper housing, and other housing-related reasons. Family-related reasons include changes in marital status, establishing one's own household, and other family-related reasons. All remaining reasons are grouped as other reasons. The data source is the 2005–2019 Current Population Survey.

3 Model

In this section, I introduce a quantitative spatial equilibrium model based on [Greaney \(2026\)](#). The model has two key components. First, households accumulate liquid and illiquid wealth, experience idiosyncratic income risk, and face life-cycle dynamics. The second component is geography and migration. Households live and work across many locations that differ in wages, housing prices, and amenities, and they face migration costs when they change their location.

This structure is useful for quantifying the value of migration. When households consider migration, their migration motives depend on their state variables. As a result, the value of migration can differ substantially across household groups. I use the model to highlight how migration value varies across these state groups and to identify the underlying mechanisms within the model.

The migration option value is determined by how much compensation households would require if migration were no longer available under the current economic environment. For this reason, I use a steady-state equilibrium without aggregate uncertainty as the benchmark for the value of migration. As a result, the model is time-invariant.

3.1 Environment

The model is set in continuous time structure. There are I discrete locations indexed by $i \in \{1, \dots, I\}$. At any given time, each household works and lives in a single location. Households observe their labor productivity (l), a location preference shock (ϵ_i), local wages (w_i), local housing prices ($p_{h,i}$), local rental housing prices ($p_{r,i}$) and amenity levels (A_i). Based on these factors, households choose a location in which to reside. Once a location is chosen, households consume a non-durable good c and housing. The consumption good is freely traded across locations, while housing is only available within each location. All local markets are perfectly competitive, and therefore there is no aggregate uncertainty in local prices.

3.2 Households Environment

Preferences Households value non-durable consumption c , housing service h , and location-specific amenities A_i . The resulting flow utility function is given by:

$$u(c, h, A_i) = \frac{(A_i c^\eta \mathbf{h}^{1-\eta})^{1-\gamma}}{1-\gamma} \quad (1)$$

where η captures the relative preference for non-durable consumption. In the model, there are two types of housing services: rental housing ($d = 0$) and owner-occupied housing ($d = 1$). Let h_r denote rental housing and h denote owner-occupied housing. One unit of rental housing provides one unit of housing services, while one unit of owner-occupied housing provides χ units of housing services:

$$\mathbf{h} = \begin{cases} h_r, & \text{if } d = 0, \\ \chi h, & \text{if } d = 1 \end{cases}$$

The parameter χ captures households' preference for homeownership. A value $\chi > 1$ implies that households have a higher preference for homeownership.

Demographics The economy is populated by finite-lived households. Age is indexed by $j \in [0, J]$. Households are born at age 0 and live up to a maximum age of J . At age j , Households face mortality risk with hazard rate $\psi(j)$. As households die, they are replaced by newborn households, so that the total population remains constant over time. At discrete age $J_s = \{0, 1, 2, \dots, J-1\}$, households experience a "shock age" event.⁶ Between shock ages, time flows continuously until households reach the next shock age. At each shock age, households are subject to idiosyncratic productivity and location preference shocks. Between shock ages, households do not experience idiosyncratic shocks. As a result, household decisions evolve deterministically between shock ages.

Earnings In labor markets, there are two types of households: working-age households and

⁶I assume that J is an integer and shock ages occur once per year.

retirees. Households of working age ($j < J_r$) supply labor inelastically with labor endowment L at local wage $w_{i,t}$. Labor endowments consist of two components: a lifecycle component $\bar{l}(j)$ and an idiosyncratic productivity component l :

$$logl = \bar{l}(j) + \ell$$

The idiosyncratic productivity ℓ follows an AR(1) process. A working-age households draws a productivity shock ϵ_j at shock ages, while a retiree does not experience productivity shocks:

$$\ell_j = \begin{cases} \theta\ell_{j-1} + \epsilon_j, & \text{if } j < J_r, \\ \ell_{j-1}, & \text{if } j \geq J_r \end{cases}, \quad j \in J_s$$

Where ϵ_j follows normal distribution ($\epsilon_j \sim N(0, \sigma^2)$). Retired households receive a pension benefit w_r , which depends on their individual labor productivity. Household earnings are therefore given by:

$$y(i, \ell, j) = \begin{cases} w_i l, & \text{if } j < J_r, \\ w_r, & \text{if } j \geq J_r \end{cases}$$

Location Choice At each shock age, households receive a "Migration shock". Upon receiving this shock, households draw idiosyncratic location preference shock $\epsilon = \{\epsilon_1, \dots, \epsilon_I\}$. The exogenous preference shocks capture household's subjective preference over locations. The preferences shocks are independently drawn from a Type I Extreme Value (Gumbel) distribution with mean 0 and scale parameter ν . The scale parameter ν governs the sensitivity of the households' location choice to idiosyncratic preference shocks. A higher value of ν implies that households place greater weight on idiosyncratic location preferences, relative to local economics characteristics such as wages, housing prices, and amenities, when choosing a location.

Once households choose a location, migrating to location i from origin i' requires paying a

non-pecuniary migration cost $\kappa_{ii'}(j)$. Migration costs vary by distance between origin and destination and age:

$$\kappa_{ii'}(j) = \kappa_0 + \kappa_1 D_{ii'} + \kappa_2 j \quad (2)$$

where $D_{ii'}$ denotes the distance between origin i and destination i' . Migration costs and location preference shocks enter the utility function in reduced form to capture their non-pecuniary nature.

This modeling approach brings a tractable framework that replicates realistic migration flows observed in the data. In particular, the model is able to match not only net migration rates but also gross in- and out-migration flows in steady state, which are important to tract households location choices.

Housing Households can obtain housing services either by renting ($d = 0$) or owning ($d = 1$). These options can not be chosen simultaneously. Rental housing size and owner-occupied housing size belong to the housing size sets H_r and H_o respectively.

At each shock age, households may adjust their tenure status and owner-occupied housing. Adjusting the owner-occupied housing state requires paying proportional transaction costs. As households buy owner-occupied housing, they must pay a fraction f_b of the new housing value. when households sell their existing owner-occupied housing, they must pay a fraction f_s of the old housing value. Renters can adjust their housing size without any transaction costs at any time. Housing depreciates at rate δ_h .

Wealth At non-shock ages ($j \notin J_s$), Households can borrow or save risk-free bonds b in the international financial market. The exogenous return on and the cost of the liquid assets is r , which is common across locations. Borrowing constraints are determined by the amount of collateral owned by households:

$$b \geq -\phi p_{h,i} h \quad (3)$$

Where ϕ captures the fraction of housing value that households can use as collateral.

Total wealth a is defined as the sum of liquid bond holdings and illiquid housing wealth:

$$a = b + p_{h,i}h$$

Since renter households do not own housing, their wealth is consist only of liquid bond holdings. Using the notation of total wealth a , the collateral constraint (3) can be rewritten as:

$$a \geq (1 - \phi)p_{h,i}h$$

At non-shock ages, households' wealth evolves according to the following budget constraint:

$$\dot{a} = (1 - \tau)(y(l, i, j) + rb) - c - p_{r,i}h_r - (\delta_h p_{h,i}h) \quad (4)$$

The first term represents after-tax income, while the second and third terms represent non-durable consumption and rental housing expenditure, respectively. The last term applies to homeowners and captures housing repair costs arising from housing depreciation. Since the model focuses on the steady state, there are no explicit capital gains from homeownership. At shock ages ($j \in J_s$), households can modify their tenure status, housing state, and location. As a result, households' wealth may change discontinuously. When households change their tenure or housing state, they must pay transaction costs equal to a fraction of their housing value. In the case of change in housing (tenure) state from h to h' , households' wealth is given by:

$$a_h(a, i, h) = \begin{cases} a, & \text{if } h' = h, \\ a + (1 - f_s)p_{h,i}h - (1 + f_b)p_{h,i}h' & \text{otherwise} \end{cases}$$

When households choose a different housing size, they must purchase housing of size h' such that their borrowing does not exceed the new borrowing constraint:

$$a_h(a, i, h) \geq (1 - \phi)p_{i,h}h'$$

When households choose a different location, homeowners must sell their housing before moving to the chosen destination. As a result, immediately after migration, all households temporarily become renters before adjusting their tenure status. Households' wealth immediately after migrating from location i to i' is given by:

$$a_m(a, i, h) = \begin{cases} a, & \text{if } i' = i, \\ a + (1 - f_S)p_{h,i}h & \text{otherwise} \end{cases}$$

3.3 Households Problem

Value Function At non-shock ages, households solve a continuous time Hamilton-Jacobi-Bellman (HJB) equation. Households' state variables are wealth a , idiosyncratic labor productivity state ℓ , location i , age j , and housing (tenure) state h . Following [Greaney \(2026\)](#), I collect the state variables into $\Omega \equiv (a, \ell, i, j, h)$. At the maximum age J , the value function satisfies $V(\Omega) = 0$ due to the absence of bequest motives. Starting from this terminal condition, the value function of finite-lived households satisfies the following HJB equation:

$$\begin{aligned} [\rho + \psi(j)]V(\Omega) &= \max_{c,h} u(c, h, A_i) + \partial_a V(\Omega) \dot{a}(\Omega; c, h) + \partial_j V(\Omega) \\ \text{s.t. } \dot{a}_t(\Omega; c, h) &\geq 0 \quad \text{if } a = (1 - \phi)p_{h,i}h \end{aligned} \tag{5}$$

The left-hand side of (5) represents the household's value function. The first term on the right-hand side is the household's maximized utility from optimal choice of non-durable good consumption c and housing h . The second term captures how changes in households wealth affect the value function through wealth accumulation or decumulation. The third and last terms reflect changes in the value function due to aging and the passage of time, respectively.

At shock ages, households make final discrete decisions regarding their tenure status or owner-occupied housing size. Let $j_s \in J_s$ denote a shock age. The value function evaluated at a shock age, after households choose the optimal housing size h^* is:

$$V_h(\Omega) = V(a(\Omega; h^*(\Omega)), \ell, i, j_s, h^*(\Omega)) \quad (6)$$

$$h^*(\Omega) = \operatorname{argmax}_{h \in H_o} V(a(\Omega; h'), \ell, i, j_s, h^*(\Omega))$$

Shock-age households choose their location prior to selecting the optimal housing size h^* . The value after choosing a new location (or remaining in the current one) is given by:

$$V_m(\Omega) = \nu \log \left(\sum_{i'} \exp (V_h(a_t^m(a, i', h), \ell, i', j, h) - \kappa_{i,i'})^{\frac{1}{\nu}} \right) \quad (7)$$

Equation (7) follows from the properties of the Type I Extreme Value (Gumbel) distribution. Appendix provides the derivation of the expected value associated with the migration choice. Households who are just at shock-ages observe idiosyncratic labor productivity shocks. The value after the productivity shock with the conditional probability density $f(\ell'|\ell)$ is given by:

$$V_l(\Omega) = \begin{cases} \int V_m(a, \ell', i, j, h) f(\ell'|\ell) d\ell', & \text{if } j < J_r, \\ V_m(\Omega) & \text{if } j \geq J_r \end{cases} \quad (8)$$

State Density The Kolmogorov Forward (KF) Equation describes the evolution of the density of households over the state space. Between shock ages, the density of state variables $g_t(\Omega)$ evolves due to households' saving behavior, aging, and mortality. The non-shock age Kolmogorov Forward (KF) equation is given by:

$$\dot{g}(\Omega) = -\partial_a \{ \dot{a}(\Omega) g(\Omega) \} - \partial_j g(\Omega) - \psi(j) g(\Omega) \quad (9)$$

In steady state, the left-hand side of equation (9) is equal to zero because there is no net change in the density of states variables. The right-hand side of equation (9) represents how the gross density of state variables evolves according to households' behavior. The first term

shows the flow of the density induced by changes in household wealth. The second and third terms capture changes in the density due to aging and mortality, respectively.

At shock ages, the first event is the realization of idiosyncratic labor productivity shocks. The density of state variables immediately after this event evolves as follows:

$$g_l(\Omega) = \begin{cases} \int g(a, \ell', i, j, h) f(\ell' | \ell) d\ell' & \text{if } j < J_r, \\ g(\Omega) & \text{if } j \geq J_r \end{cases} \quad (10)$$

The second event is the realization of idiosyncratic location preference shocks. After observing these shocks, households optimally choose their locations. The density of state variables following households' location choice is given by:

$$g_m(\Omega) = \sum_{i'} \int \mathbf{1}_m(a', i', h'; a, i, h) \mu_{i', i, t}(a', \ell, i', j, h') g_l(a', \ell, i', j, h') dF(a', h') \quad (11)$$

I define the wealth-adjusted indicator function:

$$\mathbf{1}_m(a', i', h'; a, i, h) = \begin{cases} 1 & \text{if } a_m(a', i', h') = a \text{ and } h' = h, \\ 0 & \text{otherwise} \end{cases}$$

In equation (11), $\mu_{i', i, t}(a' \ell, i', j, h')$ denotes the probability that a household with state (a', ℓ, i', j, h') migrates from location i' to i . As a result of migration, the households transitions to state (a, ℓ, i, j, h) , where post-migration wealth is determined by the migration-induced wealth adjustment. Migration probabilities follow the properties of the Type I Extreme Value (Gumbel) distribution and are given by:

$$\mu_{i'', i', t}(a', \ell, i'', j, h') = \frac{\exp(V_h(a_m(a', i', h'), \ell, i', j, h') - \kappa_{ii'}(j))^{\frac{1}{\nu}}}{\sum_{i''} \exp(V_h(a_m(a', i'', h'), \ell, i'', j, h') - \kappa_{ii''}(j))^{\frac{1}{\nu}}} \quad (12)$$

The final event of shock ages is tenure choice or owner-occupied house size choice. The

density of state variables following this choice is given by:

$$g_h(\Omega) = \sum_{i'} \int \mathbf{1}_h(a', \ell, i, j, h'; a, h) g_m(a', \ell, i, j, h') dF(a', h') \quad (13)$$

where the indicator function $\mathbf{1}_h(a', \ell, i, j, h'; a, h)$ is given by:

$$\mathbf{1}_h(a', \ell, i, j, h'; a, h) = \begin{cases} 1 & \text{if } a_h(a', i, h^*(a', \ell, i, j, h')) = a \text{ and } h^*(a', \ell, i, j, h') = h, \\ 0 & \text{otherwise} \end{cases}$$

By construction, a household with states (a', ℓ, i, j, h') transitions to states (a, ℓ, i, j, h) after making the tenure or owner-occupied house size choice.

3.4 Production

Final Good Sector In each location, there is a representative firm that produces the non-durable consumption good, which is freely traded across locations. The final good is produced according to the following neoclassical production function:

$$Y_i = Z_i L_i^\sigma \quad (14)$$

Since a representative firm produces homogeneous goods, the labor demand elasticity parameter α does not depend on location. Z_i denotes the total factor productivity (TFP) of location i . L_i is the total effective employment in location i .⁷ Considering the agglomeration effect, local productivity Z_i is given by:

$$Z_i = \bar{Z}_i L_i^\eta \quad (15)$$

where $\bar{Z}_i$ captures fixed local productivity and η represents the strength of the agglomeration effect. As effective labor supply L_i increases in location i , local productivity Z_i also increases.

⁷Since the production function exhibits decreasing return to scale, a representative firm earns positive profits. For simplicity, I assume that the profits are fully taxed by the government.

Under this setting, the equilibrium local wage in location i is given by:

$$w_i = \sigma Z_i L_i^{\sigma-1} \quad (16)$$

The total effective employment L_i is obtained by aggregating the density of state variables $g(\Omega)$ and the individual labor endowment ℓ :

$$L_i = \int \exp(\bar{\ell}(j) + \ell) g(\Omega) dF(a, \ell, j, h)$$

Construction Sector As in [Kaplan et al. \(2020\)](#), there is a representative foreign-owned firm that produces housing in each location. Housing is produced according to the following production function:

$$H_i = (X_i^H)^{1-\xi} F_i^\xi \quad (17)$$

where X^H denotes inputs of the final good, and F denotes permits. Given this production function, the unit cost of construction is given by:

$$p_{h,i} = \left(\frac{1}{1-\xi} \right)^{1-\xi} \left(\frac{p_{f,i}}{\xi} \right)^\xi \quad (18)$$

where $p_{f,i}$ is the permit price in location i . In the steady-state equilibrium, housing demand equals housing supply. Therefore, the housing price equals the unit cost of construction.

After construction, firms sell the houses to absentee landlords, who then rent the housing units to consumers. The landlords face an opportunity cost to the fixed interest rate r , and their behavior follows a no-arbitrage condition. It implies that the rental price of housing is determined as follows:

$$p_{r,i} = (r + \delta_h) p_{h,i} \quad (19)$$

3.5 Government

The government collects revenue by selling permits and taxing the positive profits of competitive firms. I assume that the government always satisfies its budget constraint. To balanced the budget, I assume that the government spends an amount G on public expenditures that do not affect household utility.

At the same time I assume that the government decides permit prices. The government's permit price rule⁸ is given by:

$$p_{f,i} = \Pi_i \left(\frac{N_i g_N}{N_{i,0}} \right)^{\beta_i} \quad (20)$$

where $N_{i,0}$ is the base year share of population in location i , g_N is aggregate population growth from the base year, and N_i is the population share of location i :

$$N_i = \int g(\Omega) dF(a, \ell, j, h)$$

I set the base year as 1970. Π_i are exogenous permit price shifters and The function also contains the house price elasticity parameter β_j . [Saiz \(2010\)](#) estimated how much a land-use regulation influences house supply elasticity that affects housing prices. I used the estimation directly to capture how land-use regulation changes a local housing price. The parameter β_j varies across locations.⁹

3.6 Equilibrium

The model includes two equilibrium prices: the local wage w_i and the local housing price $p_{h,i}$. The equilibrium wage depends on the local effective employment L_j and the equilibrium housing price depends on the local population N_i . Thus, to compute the equilibrium prices, the local labor markets must clear whereas the local housing market does not require an explicit clearing condition. This is because the model assumes that, in the steady state,

⁸The intuition behind the permit price rule follows the approach of [Saiz \(2010\)](#), who estimates the effect of local factors on housing prices. I also build on the permit price rule in [Greaney \(2026\)](#) to model permit prices.

⁹[Glaeser et al. \(2005\)](#) and [Saiz \(2010\)](#) show the empirical evidences of the US.

housing demand always equals housing supply.

I update the local wage until the local labor supply equals the local labor demand, where the local labor demand is computed from equation (16). Simultaneously, I iterate on the housing price until its value converges so that I can capture consistency with the steady-state population.

3.7 Solving the Model

The model has five state variables: wealth, idiosyncratic labor productivity, location, age, and housing status. This implies that the model has a large state space. Moreover, to reflect wealth modifications from migration choices and housing adjustment, I need to solve discrete choice problems, which requires substantial computation time even for computing the steady-state equilibrium.

To address this computational challenge, I use continuous-time dynamic programming, following Achdou et al. (2022). Continuous-time dynamic programming provides several advantages in solving the household optimization problem. First, except at the borrowing constraint, the household optimal choice can be characterized by the first-order conditions. In other words, the Euler equation holds with equality at all interior points of the state space. Moreover, the policy functions obtained from the first-order conditions require only the derivative of the current value function. This substantially reduces the computational cost of solving for optimal policies.

Second, under the continuous-time HJB equation (5), wealth evolves continuously over an infinitesimal time interval. As a result, the transition matrix associated with the HJB equation is sparse, with most of its elements equal to zero. This sparse structure greatly reduces the computational burden of finding the fixed point of the value function. In addition, the state density can be computed from the KF equation (9) by just using the transpose of the same sparse transition matrix. Therefore, once the HJB equation is solved, obtaining the stationary distribution requires relatively little additional effort.

In addition, I reflect idiosyncratic income shocks and solve migration choices, and housing adjustment decisions using the mixed-time structure of Greaney (2026). To solve the discrete-choice problems, I need to use linear interpolation over the wealth grid in order to

account for wealth modifications from migration and housing adjustment. This interpolation step is computationally costly. The mixed-time structure efficiently reduces this burden by allowing me to solve the discrete-choice problems over relatively wider time intervals within the life-cycle problem. As a result, the number of discrete-choice problems that must be solved along the path to equilibrium is substantially reduced, which significantly shortens the computation time. A more detailed description of the computational procedure is provided in Appendix E.

4 Calibration

In this section, I describe how I calibrate the model parameters to match key empirical moments. I begin by describing the data and explaining how each dataset is mapped to the model. I then discuss how the parameter values are determined. For clarity, I organize the parameters into two groups: externally calibrated parameters and internally calibrated parameters.

To map the data to the model, I use a sample of 50 Metropolitan Statistical Areas (MSAs) in the United States in 2016. I select MSAs such that their combined population sufficiently represents the overall U.S. population.

4.1 Data Description

American Community Survey I obtain geographical productivity, wages, housing prices, and migration rates from the 2016 American Community Survey (ACS), which I access through IPUMS. I first select individuals aged 20 to 90 and then exclude individuals living in group quarters. In addition, I normalize the population-weighted average house price to one across locations using person weights.

Bureau of Economic Analysis Regional Accounts The Bureau of Economic Analysis Regional Accounts (BEA) provides information on average wages and salaries, employment, and population across MSAs from 1969 to 2016. I use it to compute 2016 regional population

shares, 1970 base-year population shares, and local employment. The population shares are used to match house prices.

Survey of Consumer Finances The Survey of Consumer Finances (SCF) collects detailed information on U.S. households’ income, assets, debts, home values, and homeownership every three years. I use the 2016 SCF to measure household wealth. I define household wealth using the net worth variable and restrict the sample to households with non-negative wealth. I further restrict the sample to households aged 20 to 90 to align with the model. In all calculations, I use household weights.

4.2 Mapping Directly from the Data

Local Wages I compute local wages which determined by local final good firms using the 2016 ACS. To retain only working-age households, I restrict the sample to individuals aged 20 to 64 who are employed, worked at least 35 hours per week, and worked at least 40 weeks in the previous year. I then estimate the following regression equation:

$$\log w_n = \alpha_n + \beta_w X_n + \epsilon_n$$

where w_n is individual n ’s wage and X_n is a matrix of individual characteristics that may affect wages. I include age, age squared, a sex dummy, race dummies, and education dummies. During the estimation, I drop individuals who are self-employed, whose income is imputed, who have business income or negative income, and who have missing education data. From this regression, I average the residuals $\exp(\epsilon_n)$ within each MSA and interpret them as location-specific marginal productivity of labor. I then treat this measure of marginal productivity of labor as the local wage. Without loss of generality, I normalize these local wages so that the population-weighted average local wage is 1.

House Prices I construct MSA-level nominal house prices using the 2016 ACS. I first compute the median house value of homeowners in each MSA. I then drop top-coded and imputed house values. Without loss of generality, I normalize house prices so that the population-

weighted average house price is 1.

Migration Rates I compute migration rates using the 2016 ACS. First, I compute the number of movers who moved from a different MSA to their current MSA during the previous year. I then construct MSA-level migration rates by dividing the number of movers by the total population in each MSA.

4.3 Externally Calibrated Parameters

I borrow the model parameter values from existing literature. For some parameters, I directly compute the values from the data.

Demographics Households are living from age 20 to the maximum age $J = 80$. The mortality hazard rate $\phi(j)$ is set by using estimates from [Bell and Miller \(2005\)](#).

Life-cycle Earnings The stochastic component of the model assumes that individual productivity follows an AR(1) process in logs. The conditional persistence is set to $\theta = 0.91$, and the conditional variance is set to $\sigma^2 = 0.04$, following [Berger et al. \(2018\)](#). I follow [Hanse \(1993\)](#) to set the life-cycle component of earnings, $\bar{l}(j)$. I normalize the life-cycle component so that the median labor earnings of working-age households are 1. I choose the pension benefit w_r to match pension benefits to 40% of average labor earnings. I set the income tax rate to $\tau = 21.2\%$ based on the updated 2023 version of [McDaniel \(2007\)](#).

Housing There are two sets of parameters that determine household wealth through housing and housing prices. The first set includes the selling transaction cost f_s , the buying transaction cost f_b , and the collateral constraint ϕ . I set the two transaction costs to $f_s = 7\%$ and $f_b = 2.5\%$, following [Gruber and Martin \(2003\)](#). I also set the collateral constraint to $\phi = 0.8$, following [Piazzesi and Schneider \(2016\)](#). The other set of parameters includes the housing depreciation rate δ_h , the housing construction parameter ξ , and the long-run house price elasticity parameter β_i . I set the housing depreciation rate to $\delta_h = 2\%$, following [Piazzesi and Schneider \(2016\)](#). I set the construction parameter to $\xi = 0.38$, following

Herkenhoff et al. (2018), and set β_i following Saiz (2010)¹⁰.

Other Parameters The remaining externally calibrated parameters are as follows. I set the coefficient of relative risk aversion and the risk-free interest rate to the standard values, $\gamma = 2$ and $r = 4\%$, respectively. Following the literature, I set the elasticity of labor demand to $\alpha = 0.75$ for all locations, based on the assumption that local representative firms produce homogeneous tradable goods.

Table 1: Externally Calibrated Parameters

Parameter	Description	Value	Source / Target
Demographics			
J	Maximum age	70	Lifespan 20 - 90
J_r	Retired age	45	Retire at 65
$\psi(j)$	Mortality hazard rate		Bell and Miller (2005)
Life-cycle Earnings			
θ	Conditional persistence	0.91	Berger et al. (2018)
σ^2	Conditional variance	0.04	Berger et al. (2018)
$\bar{l}(j)$	Life-cycle component of earnings		Hanse (1993)
w_r	Pension benefit		40% of average labor earnings
τ	Income tax rate	21.2%	McDaniel (2007) (ver. 2023)
Housing			
f_s	Selling transaction cost	7%	Gruber and Martin (2003)
f_b	Buying transaction cost	2.5%	Gruber and Martin (2003)
ϕ	Collateral constraint	0.8	Piazzesi and Schneider (2016)
δ_h	Housing depreciation rate	2%	Piazzesi and Schneider (2016)
ξ	Housing construction parameter	0.38	Herkenhoff et al. (2018)
β_i	Long-run house price elasticity	See table	Saiz (2010)
Other Parameters			
γ	Coefficient of relative risk aversion	2	Standard value
r	Risk-free interest rate	4%	Standard value
α	Elasticity of labor demand	0.75	Literature

Note: This table summarizes externally calibrated parameters. The time unit for demographics is one year. Parameters are either taken from the existing literature, directly computed from the data, or set to standard values.

¹⁰I use the estimates from the second column of Table 5 and obtain the price elasticity by inverting the estimates.

4.4 Internally calibrated parameters

In this section, I introduce the parameters that are internally calibrated within the model. These parameters fall into four categories: preferences, house size, migration related, and local fundamentals.

Preference I choose the discount rate ρ to match the median wealth to income ratio (2.24) in the 2016 SCF. The owner-occupied house premium χ is chosen to match the homeownership rate (69.5%) in the 2016 SCF. The non-durable good preference weight α is chosen to match the housing expenditure share of 21% from [Piazzesi and Schneider \(2016\)](#).

House Sizes To internally calibrate the housing-related parameters, I follow the calibration strategy in [Kaplan et al. \(2020\)](#). First, to construct the owner-occupied housing grid $\mathbf{H}_O = \{h_1, h_2, h_3, h_4, h_5, h_6\}$, I choose the minimum housing size, maximum housing size, and the number of nodes. The owner-occupied housing grid is constructed as an evenly spaced grid. These parameter values are chosen to match the 30%, 50%, and 70% moments of the homevalue to wealth ratio from the 2016 SCF. Next, I construct the rental housing grid $\mathbf{H}_R = \{h_{r,1}, h_{r,2}, h_{r,3}\}$. Following [Kaplan et al. \(2020\)](#), I set the number of rental housing grid nodes to one-half of the number of owner-occupied housing grid nodes. The rental housing grid is also constructed as an evenly spaced grid. I then calibrate the minimum and maximum rental housing sizes so that the average housing service size of homeowners is 1.5 times larger than that of renters.

Migration There are three parameters in the linear migration cost equation (2). The intercept κ_0 is chosen to match the overall migration rate (2.53%). The distance-related cost coefficient κ_1 is chosen to match the correlation between distance and the out-migration rate between two MSAs (-0.243). For this target, I use the distance data provided by [Gardner and Hendrickson \(2018\)](#) and normalize distance by the minimum distance. The last parameter is κ_2 , which captures how age affects migration costs. I target the migration rate gap between working-age households (2.82%), and retirees (1.26%).

These two migration-cost-related parameters are reasonable to include. For example, when two locations are farther apart, the psychological cost of migration is naturally higher because migrants are more likely to face disconnected social networks and unfamiliar environments. At the same time, older households are less likely to move to other areas because they tend to have deeper familiarity with their current location, stronger family ties, and owner-occupied housing, which involves higher pecuniary moving costs due to transaction costs.

The migration elasticity ν is chosen to target the log variance of local wages (1.059). The scale parameter of the Gumbel distribution determines how sensitively households respond to random preference shocks. A higher value of ν implies that households are more sensitive to idiosyncratic local preference factors, rather than to local wages, housing prices, and amenities. Thus, the scale parameter can be interpreted as migration elasticity. When ν is high, migration is driven more by random preference shocks, so local wages do not differ much across locations. In contrast, when ν is low, households sort more strongly across locations based on economic fundamentals, which generates larger differences in local wages.

Local Fundamentals I calibrate local fundamentals using local wages, house prices, and population shares. The local productivity Z_i , permit price shifter Π_i , and local amenity parameter A_i are internally calibrated so that the model matches the corresponding empirical moments.

First, I update local productivity Z_i until model wages closely match wages from the 2016 ACS. Second, I update the permit price shifter Π_i until model house prices closely match house prices from the 2016 ACS. Lastly, I update the local amenity parameter A_i until model population shares closely match population shares from the BEA data.

However, calibrating local productivity Z_i requires one additional step. I first calibrate the agglomeration parameter η before updating Z_i . As shown in equation (15), local productivity Z_i depends on effective labor employment L_i and the agglomeration parameter η . I start with an initial guess for the agglomeration parameter η . Given this guess, I update local productivity Z_i until model wages match the empirical wages. I then compute the model-generated correlation between local productivity and the in-migration share and compare it

with the empirical target (0.838)¹¹. If the model moment does not match the target, I update the guess for η and repeat the procedure. I continue this process until the model-generated correlation is sufficiently close to the empirical target.

Table 2: Internally Calibrated Parameters

Parameter	Description	Value	Source / Target
Preferences			
ρ	Discount rate	0.036	Median wealth to income (2.24)
χ	Ownership premium	1.003	Homeownership rate (69.5%)
α	Non-durable good preference	0.79	Housing expenditure share (21%)
House Size			
h_1	Min owned house size	1.50	Home value to wealth ratio
h_6	Max owned house size	6.90	Home value to wealth ratio
$h_{r,1}$	Min rented house size	1.30	Relative owned house size (1.5)
$h_{r,3}$	Max rented house size	4.15	Relative owned house size (1.5)
Migration			
κ_0	Migration cost	9.38	Overall migration rate (2.53%)
κ_1	Migration cost	0.0143	Corr(distance, out-migration)
κ_2	Migration cost	0.055	Working-retiree migration gap (1.56%)
ν	Migration elasticity	1.520	Log variance of wages (1.059)
Local Fundamentals			
Z_i	Local productivity	See table	Local wages
Π_i	Permit price shifter	See table	House prices
A_i	Amenities	See table	Population
η	Agglomeration	0.13	Corr(productivity, in-migration)

Note: This table summarizes the internally calibrated parameters. Each parameter is chosen within the model to match the corresponding target moment. As described in the main text, the owner-occupied housing grid has 6 points, and the rental housing grid has 3 points.

4.5 Untargeted Moments

In this section, I present untargeted moments to evaluate how well the model matches the data. I examine the wealth to income ratio, house value to wealth ratio, migration rates,

¹¹Local productivity is backed out by inverting the wage equation (16). Specifically, after constructing MSA-level residual wages from the 2016 ACS and combining them with BEA employment data, local productivity is calculated as

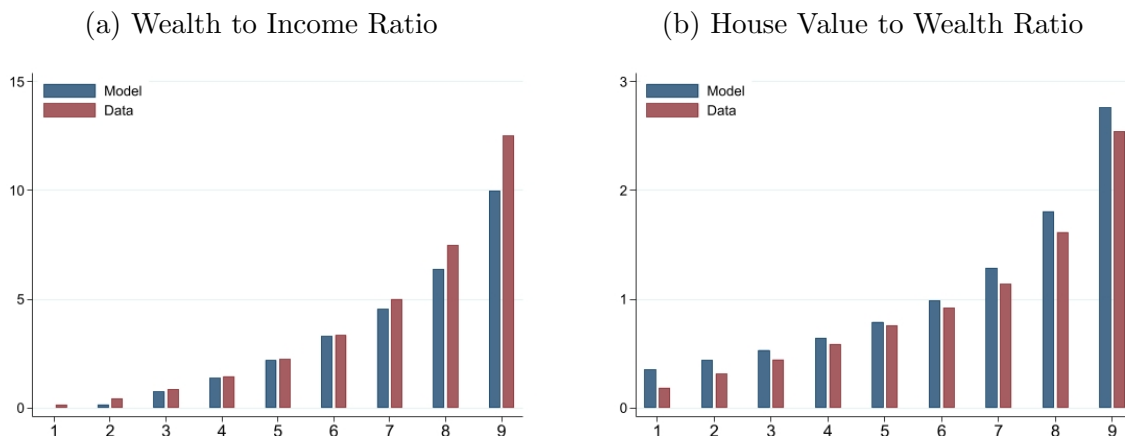
$$Z_i = \frac{W_i}{\alpha L_i^{\alpha-1+\eta}},$$

The productivity measure is then normalized so that the population-weighted average productivity equals one.

homeownership rates, and in-migration patterns. These moments are central to the model because the value of migration can be strongly affected by migration patterns, tenure status, and the amount of wealth that households have accumulated.

Wealth and House Value Distribution. Figure 4 plots the distributions of wealth and house value in the data and the model. I target the 50th percentile of the wealth to income ratio and the 30th, 50th, and 70th percentiles of the house value to wealth ratio. Figure 4a shows that the model matches the overall wealth distribution well up to the 80th percentile, while it underpredicts wealth accumulation in the top decile. Figure 4b shows that the model also matches the house value to wealth ratio well across most deciles. Again, the discrepancy gradually increases and is largest in the top decile. However, since the migration decisions of high wealth households are less sensitive to individual conditions, this distributional discrepancy is not a major concern for the migration value analysis.

Figure 4: Distributions of Wealth and Housing Wealth

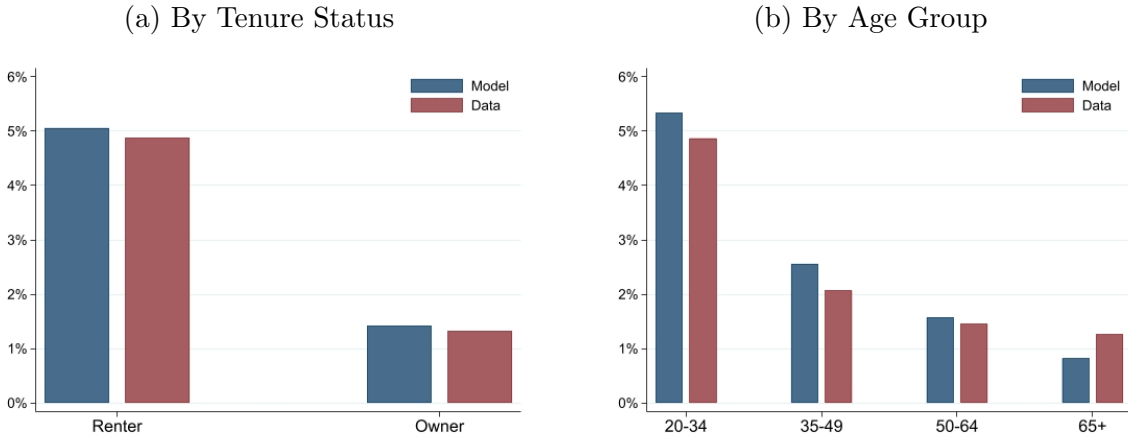


Notes: These figures compare the distributions of the wealth to income ratio and the housing wealth to total wealth ratio in the calibrated model and the data. The data source is the 2016 Survey of Consumer Finances.

Migration Rates Figure 5 plots the annual migration rates across demographic groups in the calibrated model and the data. The model is calibrated to match the overall migration rate and the migration rate gap between working-age and retired households using non-pecuniary migration costs. Figure 5a shows how well the model matches migration rates by tenure status. In both the calibrated model and the data, renters have higher annual

migration rates than owners. At the same time, the migration rate gap between the model and the data is very small. In the model, renters' and owners' migration rates are 5.05% and 1.42%, respectively. In the data, the migration rates for renters and owners are 4.88% and 1.32%, respectively. Figure 5b shows how well the model matches migration rates by age group. Since the model includes migration costs related to household age, it replicates the gradual decline in migration rates as households get older. The overall migration rate gap is around 0.5%, while the smallest gap is 0.1% for ages 50–64.

Figure 5: Migration Rates by Demographic Groups



Notes: These figures compare the annual migration rates by tenure status and age group in the calibrated model and the data. The data source is the 2016 American Community Survey.

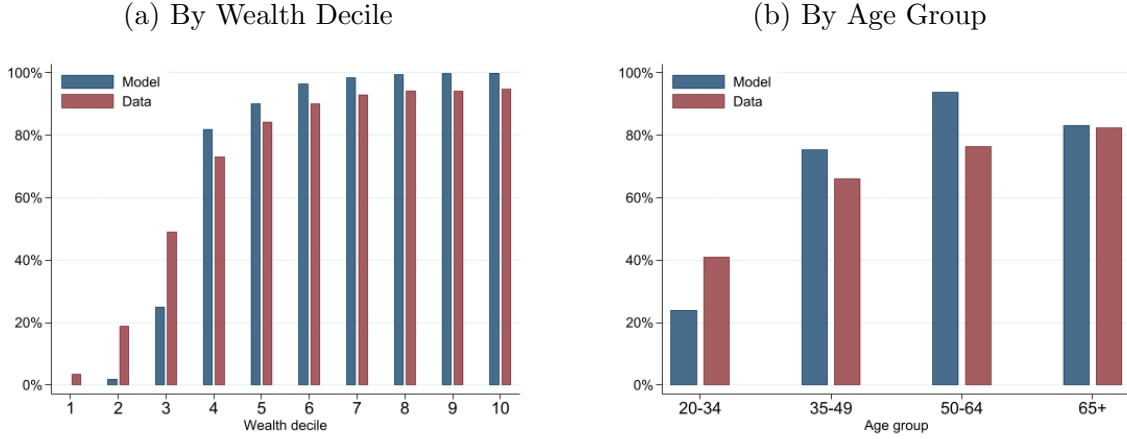
Homeownership Rates The homeownership rate affects households' migration rates because homeowners must pay transaction costs when they sell their houses to move to another location.¹² At the same time, wealthier and older households are more likely to be homeowners, which reduces their migration rates. Thus, matching homeownership rates in the data is important for validating migration rates in the model.

Figure 6 shows how homeownership rates in the model differ from those in the data. Figure 6a shows how homeownership rates deviate from the data moments across wealth deciles. After the 4th decile, the model matches the data well. From the 1st to the 4th decile, homeownership rates are lower than the data moments due to the assumption that all working-age

¹²Recall that when homeowners move to another location, they must first sell their house and become renters in the destination location before adjusting their housing size. After migration, homeowners' adjusted wealth is given by $a_m(a, i, h)$.

households are born with no wealth. Figure 6b shows homeownership rates across age groups. The model reflects the tendency that older households are more likely to be homeowners, especially in the 65+ group. The largest homeownership rate gap is for the youngest group, where the model moment is 23.9% while the data moment is 41%.

Figure 6: Homeownership Rates by Wealth and Age Groups



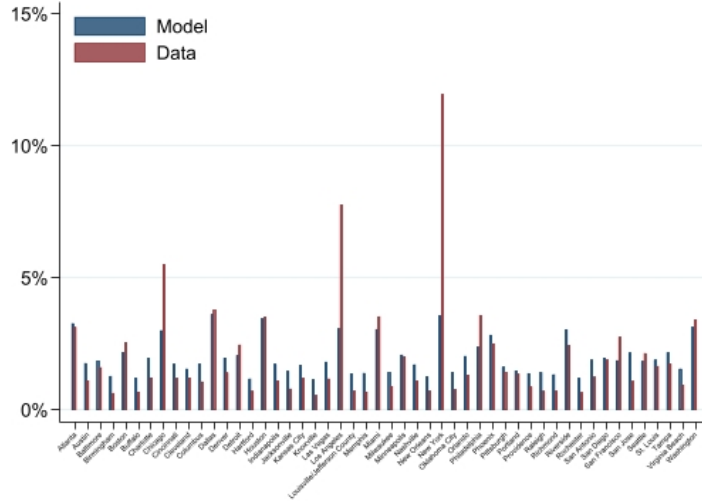
Notes: These figures compare the homeownership rates by wealth level and age group in the calibrated model and the data. The data source is the 2016 Survey of Consumer Finances. I use net worth variable as a wealth level in the data.

In-Migration Patterns I analyze where households move. Figure 7 displays the in-migration share of movers across MSAs. A few large MSAs attract a very high share of movers. This may be because the model does not include sufficiently rich sources of heterogeneity to fully explain why households move to particular locations. The model includes homogeneous idiosyncratic preference shocks and amenity parameters that are calibrated to target population shares across MSAs. However, the overall shape of the in-migration share is similar in both the model and the data. This suggests that the calibrated model captures the broad patterns of in-migration reasonably well.

To provide additional evidence on how well the calibrated model reflects in-migration patterns, I compute the correlations between in-migration shares and local characteristics: productivity,¹³ wages, and housing prices. These correlations provide an additional validation of the model's ability to reproduce destination choices across locations. Table 3 shows

¹³The correlation between the in-migration share and local productivity is a target moment used to calibrate the agglomeration parameter η .

Figure 7: In-Migration by MSA



Notes: This figure compares the in-migration shares of movers across MSAs in the calibrated model and the data. The data source is the 2016 American Community Survey.

that the in-migration share is strongly correlated with local productivity (0.81 and 0.84 in the model and the data, respectively), local wages (0.44 and 0.56 in the model and the data, respectively), and housing prices (0.40 and 0.54 in the model and the data, respectively). Although the correlations in the model and the data are not very close to each other, the calibrated model reasonably well reflects households' migration motives related to local characteristics.

Table 3: In-Migration Share and Location Characteristics

Characteristics	Model	Data
Productivity	0.81	0.84
Wages	0.44	0.56
Housing prices	0.40	0.54

Note: This table reports the correlation between the in-migration share and the destination characteristics. The data source is 2016 merican Community Survey.

5 Quantitative Analysis

In this section, I evaluate the value of migration under the baseline setting. To measure the migration value, I shut down the migration option in the steady-state equilibrium. I then focus on the mechanisms that generate different migration values across household groups by wealth, age, and tenure status. Lastly, I conduct a counterfactual experiment to evaluate how the value of migration differs across environments.

5.1 Measuring Migration Value

I measure the value of migration by computing the Consumption Equivalent Variation (CEV). The value of migration is defined as the additional consumption required to make households indifferent between the baseline model with migration and the counterfactual model without migration. In other words, it is the welfare gain from the migration option. Thus, the migration value is exactly equivalent to the CEV. Given the state variables Ω , the option value of migration $\omega(\Omega)$ is given by:

$$\omega(\Omega) = \left(\frac{V_o(\Omega)}{V'_o(\Omega)} \right)^{\frac{1}{1-\gamma}} - 1 \quad (21)$$

where $V_o(\Omega)$ denotes the value in the baseline model and $V'_o(\Omega)$ denotes the value in the model without migration. I compute $V_o(\Omega)$ and $V'_o(\Omega)$ by integrating the value function $V(\Omega)$ over the state distribution $g(\Omega)$ in each respective model. To compare the same household groups, I fix the group-specific state density from the baseline model and use this density when averaging the CEV across households within each group. In the Appendix, I show how I derive equation (21).

5.2 Quantifying the Value of Migration

In this section, I report the value of migration and migration rates across household groups. To compute the migration value, I shut down the migration option in the baseline steady

state. Specifically, I remove equations (7) and (11) from the baseline model ¹⁴. Then, using equation (21), I compare welfare between the baseline model and the counterfactual model without the migration option for the same household groups.

This CEV captures two different components of the value of migration. First, it captures the current value of migration. Households can choose to move to another location and obtain additional utility today. Second, the value function $V(\Omega)$ includes all possible future values that households can obtain over their lifetime. Therefore, shutting down the migration option also removes all potential value gains from future migration.

Migration Value Table 4 reports the size of the migration value across household groups. Overall, the aggregate migration value is 2.56%. This means that, on average, households would need 2.56% more consumption in the economy without migration to be indifferent between having and not having the migration option. However, this aggregate number masks substantial heterogeneity across households. The migration value is especially high for young and wealthy households, while it is much smaller for older and low-wealth households. Therefore, the migration option does not provide the same value to all households. Instead, its value depends strongly on both life-cycle position and wealth.

To reflect this heterogeneity, I first divide households into 16 groups by wealth quartile and four age groups. I then divide households into 8 groups by tenure status and the same age groups. The migration values in Table 4 clearly differ across household groups.

The first clear pattern is that the migration value decreases with age. For all households, the migration value is 3.87% for households aged 20–34, but it declines to 1.25% for households aged 65 and above. This pattern appears within almost all wealth and tenure groups. This result suggests that younger households benefit more from having the migration option, because they have a longer remaining lifetime over which migration can affect future utility. In contrast, older households have fewer future periods to use the migration option and enjoy the welfare gain from the migration, so the lifetime value of migration is smaller. In addition, migration costs increase with age following equation (2), older households face more costly

¹⁴By excluding equation (7), I remove households' value gains from migration. By excluding equation (11), I remove the reallocation of household mass across locations through migration choices.

migration opportunities over their remaining lifetime, which further reduces the value of the migration option.

The second pattern is that the migration value generally increases with wealth within each age group. For example, among households aged 20–34, the migration value increases from 3.71% for the lowest wealth quartile to 5.96% for the highest wealth quartile. A similar pattern also appears among households aged 35–49, 50–64, and 65 and above. This implies that wealthier households obtain larger value from the migration option, conditional on age. The tenure-status results show that renters and owners have similar migration values among young households, but the difference becomes larger at older ages. For households aged 20–34, renters and owners have almost the same migration value, 3.87% and 3.86%, respectively. However, among households aged 65 and above, the migration value is 0.83% for renters and 1.32% for owners.¹⁵ This suggests that homeownership is more closely related to the value of migration at older ages, while the difference between renters and owners is small for younger households.

Migration Rate Table 5 reports migration rates across household groups. Several patterns emerge from the table. First, migration rates decline with age, from 4.88% among households aged 20–34 to 1.17% among households aged 65 and above. Second, migration rates generally decline with wealth among households below age 65. For example, among households aged 20–34, the migration rate falls from 5.31% in Q1 to 3.74% in Q4. Third, renters migrate more frequently than homeowners. The aggregate migration rate is 4.30% for renters but only 1.79% for owners.

Ownership is one useful way to understand these migration rate patterns. Homeowners face not only the utility cost of migration but also pecuniary transaction costs¹⁶. Therefore, higher ownership rates can reduce migration rates. This helps explain why renters have higher mi-

¹⁵Although homeowners have larger migration values than renters within most age groups after age 35, the aggregate migration value is higher for renters. This pattern comes from differences in the age composition of tenure groups. Renters are more concentrated among young households, and young households have higher migration values. In contrast, homeowners are more concentrated among older households, whose migration values are relatively low. As a result, when I compute the population-weighted aggregate value, renters have a higher migration value than homeowners.

¹⁶Recall that when homeowners move to another location, they must pay a transaction cost, f_s , to sell their current house.

Table 4: Migration Value by Group

Group	20–34	35–49	50–64	65+	Aggregate
Wealth quartile by age					
Q1	3.71%	2.25%	1.60%	0.57%	2.69%
Q2	3.88%	2.57%	1.81%	0.85%	2.26%
Q3	4.56%	3.08%	2.01%	1.21%	2.22%
Q4	5.96%	4.32%	2.92%	1.97%	2.96%
Tenure status by age					
Renter	3.87%	2.48%	2.37%	0.83%	3.04%
Owner	3.86%	3.15%	2.38%	1.32%	2.34%
Aggregate households by age					
Aggregate	3.87%	2.99%	2.38%	1.25%	2.56%

Note: This table reports the Consumption Equivalent Variation (CEV) from migration option, holding prices fixed at the baseline level.

gration rates than homeowners. Table 6 shows that ownership rates increase strongly with both wealth and age. For example, among households aged 20–34, the ownership rate rises from 6.74% in Q1 to 99.01% in Q4. Within the lowest wealth quartile, the ownership rate also increases from 6.74% for households aged 20–34 to 58.45% for households aged 65 and above. Since owners are less mobile than renters, this pattern helps explain why migration rates tend to decline with wealth and age.

However, ownership alone cannot fully explain the migration rate patterns. While migration rates decline with wealth among young households, the pattern becomes less monotonic at older ages. For example, among households aged 50–64, migration rates decline from Q1 to Q2 but then increase toward Q4. Among households aged 65 and above, migration rates increase from 0.98% in Q1 to 1.34% in Q4 despite higher ownership rates among wealthier households. Thus, to fully explain the variation in migration rates across age and wealth groups, I discuss the main economic intuition behind these patterns below.

The intuition behind the age gradient in migration rates is similar to the migration value gradient. Migration costs increase with age following equation (2). In addition, older households have a shorter remaining lifetime over which they can benefit from relocation. As the

remaining horizon shrinks, households have fewer opportunities to exercise the migration option in the future and less time to enjoy the gains from moving. Consequently, the expected net benefits of migration decline with age, reducing the incentive to incur the utility cost of moving and leading to lower migration rates among older households.¹⁷

The wealth gradient in migration rates can be understood through differences in the need to use migration as a tool for improving welfare. Wealthier households can smooth consumption and housing expenditures more easily without relocating because they possess more accumulated wealth. In contrast, low-wealth households that face binding borrowing constraints have stronger incentives to improve their economic situation through migration. By moving to locations with better earning opportunities, they can accumulate more wealth over time. Alternatively, they may relocate to areas with lower living costs, allowing them to increase current consumption and saving. Consequently, migration is more frequently used by low-wealth households as a tool for improving welfare, while wealthy households move only in selected states where relocation brings large gains.

The interaction between age and wealth reflects differences in migration motives across household groups. Since, young low-wealth households depend more heavily on wages, so their migration incentives are closely related to future gains from better earning opportunities.¹⁸ In contrast, high-wealth households depend less on wages and are therefore less interested in migration for future gains through higher labor income. Older households also have shorter remaining lifetimes, so they place less weight on future gains from migration. This implies that migration motives become increasingly related to current gains¹⁹ rather than future gains for wealthier and older households. For older low-wealth households, however, limited wealth makes it difficult to generate large current utility gains through consumption

¹⁷Age does not enter the migration probability directly in equation (12). However, migration probabilities depend on the age-specific value function V_h . Since the HJB is solved backward from the terminal age, the value function of older households reflects a shorter remaining lifetime. Consequently, the continuation-value gain from relocating is generally smaller for older households, reducing the migration surplus and lowering migration probabilities.

¹⁸Table 7 in the next section shows that young and low-wealth households are more likely to move to highly productive locations, while older or wealthier households are more likely to move to less expensive locations.

¹⁹Although moving to a larger house allows households to enjoy higher housing consumption in future periods, it is not a form of migration motivated by sacrificing current consumption in exchange for better future opportunities. For this reason, I classify housing-related gains as current gains rather than future gains.

or housing adjustments. As a result, migration rates decline more rapidly for low-wealth households. High-wealth households, by contrast, have lower migration rates at younger ages because their migration incentives are weaker to begin with. However, their migration incentives decline more slowly with age because they have enough wealth to obtain large current gains from relocation.²⁰ This helps explain why migration rates can be higher for wealthier households within older age groups.

Table 5: Migration Rate by Group

Group	20–34	35–49	50–64	65+	Aggregate
Wealth quartile by age					
Q1	5.31%	2.71%	1.80%	0.98%	3.84%
Q2	4.09%	2.45%	1.73%	1.07%	2.38%
Q3	3.77%	2.39%	1.76%	1.17%	1.90%
Q4	3.74%	2.64%	1.91%	1.34%	1.91%
Tenure status by age					
Renter	5.35%	3.19%	2.40%	1.17%	4.3%
Owner	3.02%	2.30%	1.79%	1.17%	1.79%
Aggregate households by age					
Aggregate	4.88%	2.54%	1.83%	1.17%	2.55%

Note: This table reports the baseline migration rate by wealth, tenure, and age group.

Value vs Rate Comparing Tables 4 and 5 reveals both similarities and important differences between migration values and migration rates. Across age groups, the two measures display similar patterns, as households with larger expected gains from relocation tend to exhibit both higher migration rates and larger migration values. Younger households migrate more frequently and receive larger benefits from having the migration option, whereas both migration rates and migration values decline with age. As discussed above, as households age, migration costs increase. In addition, because older households have shorter remaining lifetimes, they have fewer opportunities to migrate and less time to enjoy the value generated

²⁰Households aged 65 and above receive pension benefits that do not vary across locations. Thus, they do not have an investment motive from wage gains through migration. Migration decisions for this group are driven by current gains, such as increases in current consumption or housing expenditures.

Table 6: Ownership Rate by Group

Group	20–34	35–49	50–64	65+
Wealth quartile by age				
Q1	6.74%	28.24%	46.60%	58.45%
Q2	46.99%	81.51%	92.12%	75.58%
Q3	89.63%	98.67%	99.26%	94.64%
Q4	99.01%	99.88%	99.87%	99.40%

Note: This table reports the ownership rate by wealth and age group.

by relocation. Thus, both migration rates and migration values decline with age. ²¹ However, the relationship differs across wealth groups. Migration values increase with wealth, but migration rates do not follow the same pattern. Among households aged 20–34, migration rates decline as wealth increases, while the relationship becomes less clear at older ages. A similar discrepancy appears for tenure groups. Homeowners migrate less frequently than renters, but their migration values are similar or even larger than those of renters in several age groups. These patterns show that the distribution of migration values cannot be explained only by the mechanism that generates changes in migration rates. To understand why migration values differ across wealth groups, I next examine why households move and what gains they obtain from relocation.

5.3 Migration Motives

In this section, I analyze the migration motives of different household groups. As explained in the previous section, identifying why households move and where they relocate is important for understanding the migration value.

Where do households go? Migration motives in the model are closely related to wages and housing prices. Households can increase current and future utility by moving to locations with higher wages or lower housing prices. Hence, examining the wage and housing price changes experienced by movers helps identify the motives behind migration decisions across

²¹The effect of the remaining lifetime on migration value is incorporated through equation (7).

household groups.

Table 7 reports the average changes in destination wages and housing prices relative to the origin location for each household group. A clear pattern emerges for young households in the lowest wealth quartiles. Both groups move toward locations with higher wages and higher housing prices. For example, households aged 20–34 in the lowest wealth quartile experience a wage increase of 3.93% and a housing price increase of 9.35% after moving.²² Similar patterns appear for young households in the second wealth quartile. This behavior is consistent with an investment motive. Young low-wealth households are willing to bear migration costs and move to more expensive locations because higher wages improve future earning opportunities. Although current housing expenditures increase, the higher future income allows these households to accumulate more wealth over their lifetime. As a result, relocating to a location with better labor market opportunities can increase lifetime utility despite the higher current cost. Households aged 35–49 show a similar but weaker pattern. This suggests that the investment motive still exists for this group, but it is weaker than for younger households due to the shorter remaining lifetime.

The remaining household groups display a very different pattern. Most of them move toward locations with both lower wages and lower housing prices. This suggests that their migration decisions are driven by motives other than investment in future earnings. One possible explanation is a reducing living cost motive. By relocating to areas with lower housing prices, households can reduce housing expenditures and allocate a larger share of resources to non-housing consumption, which represents a larger component of utility in the model. Another possibility is a housing adjustment motive. Lower housing prices allow households to purchase or rent larger housing units, generating additional utility through higher housing consumption. Both motives generate gains through an immediate improvement in current utility rather than through future wealth accumulation. However, it is not yet clear which household groups relocate primarily to reduce housing costs and which relocate to adjust their housing consumption. Thus, additional analysis is required to identify the motives behind the observed tendency of households to move toward locations with lower housing

²²Even though the housing price increases, the corresponding user cost rises by a smaller amount. For renters, the housing cost is given by equation (19). For homeowners, the user cost is given by $((1-\tau)r+\delta_h)p_{h,i}$.

prices.

Table 7: Mover Destination Wage and Housing Price Changes by Group

	Age Group			
	20–34	35–49	50–64	65+
Panel A. Wage Change				
<i>Wealth quartile by age</i>				
Q1	3.93%	2.15%	-5.17%	0.00%
Q2	0.81%	-0.06%	-7.87%	0.00%
Q3	-1.58%	-3.31%	-8.43%	0.00%
Q4	-6.40%	-6.84%	-9.21%	0.00%
<i>Tenure status by age</i>				
Renter	3.68%	2.79%	-8.29%	0.00%
Owner	-3.18%	-4.11%	-8.48%	0.00%
Panel B. Housing Price Change				
<i>Wealth quartile by age</i>				
Q1	9.35%	7.98%	-17.13%	-27.66%
Q2	4.24%	2.59%	-24.18%	-31.93%
Q3	-3.47%	-7.29%	-23.71%	-31.14%
Q4	-16.59%	-17.85%	-27.40%	-31.59%
<i>Tenure status by age</i>				
Renter	9.18%	9.72%	-28.96%	-38.58%
Owner	-7.22%	-9.46%	-24.61%	-30.01%

Notes: Panel A reports the average percentage change in wages between movers’ destination and origin locations. Positive values indicate that movers relocate to locations with higher wages, while negative values indicate that movers relocate to locations with lower wages. For households aged 65 and above, wage changes are zero because their income is given by pension benefits, which do not vary across locations. Panel B reports the average percentage change in housing prices between movers’ destination and origin locations. Positive values indicate that movers relocate to locations with more expensive housing markets, while negative values indicate that movers relocate to locations with less expensive housing markets.

Households’ Consumption Changes Table 8 reports changes in non-durable good consumption and housing size after migration. Several patterns emerge from the tables. Among households aged 20–34, consumption gains decline with wealth, ranging from 4.97% for the lowest wealth quartile to -6.54% for the highest wealth quartile. By contrast, housing size gains increase strongly with wealth, rising from -6.78% to 23.33%. Similar patterns appear for households aged 35–49 and 50–64. For example, among households aged 50–64, consump-

tion changes fall from 0.83% for Q1 to -3.95% for Q4, while housing size gains increase from 2.24% to 24.66%. The tenure groups display a similar pattern. Renters experience larger consumption gains than owners, whereas owners tend to obtain larger housing size gains.

The consumption changes across wealth groups help explain why migration values are higher for high-wealth households. Young low-wealth households mainly move toward productive locations and increase non-durable consumption through higher labor income because non-durable consumption receives a relatively large weight in utility. At the same time, the high housing prices in productive locations often lead to decline in housing consumption.

As low-wealth households become older, their migration motives change. Rather than moving to productive locations, they increasingly relocate to areas with lower living costs. As a result, they are able to increase both non-durable consumption and housing consumption through migration. However, migration values for low-wealth households remain smaller than those of high-wealth households because their limited wealth restricts the size of housing and consumption gains from relocation.

High-wealth households obtain migration gains mainly by increasing housing size after relocation. This is consistent with the fact that these households are more likely to already consume relatively high levels of non-durable goods. Since the marginal utility of non-durable consumption is lower at these consumption levels, the utility loss from giving up some non-durable consumption is relatively small. In contrast, housing size choices are discrete in the model. Therefore, if households can obtain a large housing size increase after moving to a lower-cost location, the utility gain from this discrete housing adjustment can be large. This mechanism helps understand why migration values tend to increase with wealth. Low-wealth households start from relatively low levels of consumption and housing services, so a reduction in housing size can generate a large marginal utility loss. At the same time, for middle-aged households, limited wealth restricts the size of the consumption gains they can obtain from relocation.²³ In contrast, high-wealth households can generate large migration gains through substantial increases in housing size because their higher wealth allows them

²³Most Q1 households are renters. Since the maximum rental housing size is smaller than the maximum owner-occupied housing size, their potential housing-size increase is limited. Moreover, even if they choose to become homeowners, the housing-size increase remains limited because the buying transaction cost reduces the resources available for housing adjustment.

to make larger housing adjustments.

However, the relationship becomes more ambiguous for households aged 65 and older. The last column of Table 8 shows that, with the exception of Q4 households, the increases in both non-durable consumption and housing consumption become smaller as wealth rises. As a result, it becomes less clear why migration values continue to increase with wealth among older households. I therefore view this pattern among elderly households as a limitation of the current model and leave it as a topic for future research.

Table 8: Mover Consumption and Housing Size Changes by Group

	Age Group			
	20–34	35–49	50–64	65+
Panel A. Consumption Change				
<i>Wealth quartile by age</i>				
Q1	4.97%	2.16%	0.83%	11.96%
Q2	-0.03%	-0.90%	-1.07%	4.22%
Q3	-2.81%	-3.78%	-2.76%	2.53%
Q4	-6.54%	-5.97%	-3.95%	-0.09%
<i>Tenure status by age</i>				
Renter	4.37%	2.16%	-1.42%	3.68%
Owner	-3.77%	-3.98%	-2.96%	3.23%
Panel B. Housing Size Change				
<i>Wealth quartile by age</i>				
Q1	-6.78%	-7.69%	2.24%	14.13%
Q2	-3.47%	-5.23%	5.31%	11.81%
Q3	6.57%	7.37%	11.98%	8.59%
Q4	23.33%	22.06%	24.66%	20.98%
<i>Tenure status by age</i>				
Renter	-6.46%	-7.88%	14.52%	23.47%
Owner	7.04%	9.14%	16.76%	12.93%

Notes: Panel A reports movers' average non-durable consumption changes upon moving. Panel B reports movers' average housing size changes upon moving.

5.4 Environmental Effect

In this section, I examine how changes in the economic environment affect migration values and migration rates. Specifically, I study how changes in the dispersion of local wages and housing prices alter migration behavior and the distribution of migration gains across household groups. Since migration is fundamentally driven by differences in wages and housing prices across locations,²⁴ the spatial distribution of wages and housing prices plays an important role in shaping both migration patterns and the value of migration.

To investigate this effect, I conduct a counterfactual experiment in which I increase the dispersion of local productivity by one standard deviation while keeping average productivity constant. Given the new productivity distribution, I solve for a new general equilibrium and recompute household decisions, migration patterns, and migration values across household groups.²⁵ Because long-run changes in the economic environment also alter the composition of household groups, I do not hold group composition fixed at its baseline steady-state distribution.

Table 9 reports how the distribution of local wages and housing prices changes relative to the baseline economy under the productivity-distribution counterfactual. The results are clear: both wages and housing prices gap among locations become larger. The standard deviation of local wages increases by 0.13, while the standard deviation of housing prices increases by 0.16. The mean values of both wages and housing prices also increase. This occurs because, in productive cities, agglomeration effects and high housing price elasticities amplify the effects of population inflows and housing demand, generating nonlinear increases in wages and housing prices. This result can also be interpreted as a tight migration barrier to productive cities. Figure 8 displays how population shares change in the counterfactual economy relative to the baseline economy across cities ordered by productivity levels. The figure shows that, as productivity becomes more dispersed, households tend to concentrate

²⁴Recall that the investment motive is represented by migration toward higher-wage locations, whereas the living-cost reduction motive is represented by migration toward lower-housing-price locations.

²⁵Such changes in productivity dispersion can be interpreted as situations in which regional inequality increases across locations, for example when highly productive cities such as San Francisco continue to prosper while less productive cities such as Detroit experience relative decline. However, in this counterfactual, I hold average productivity constant and increase only the dispersion of productivity across locations. By doing so, I isolate the effect of local growth on migration behavior and migration values.

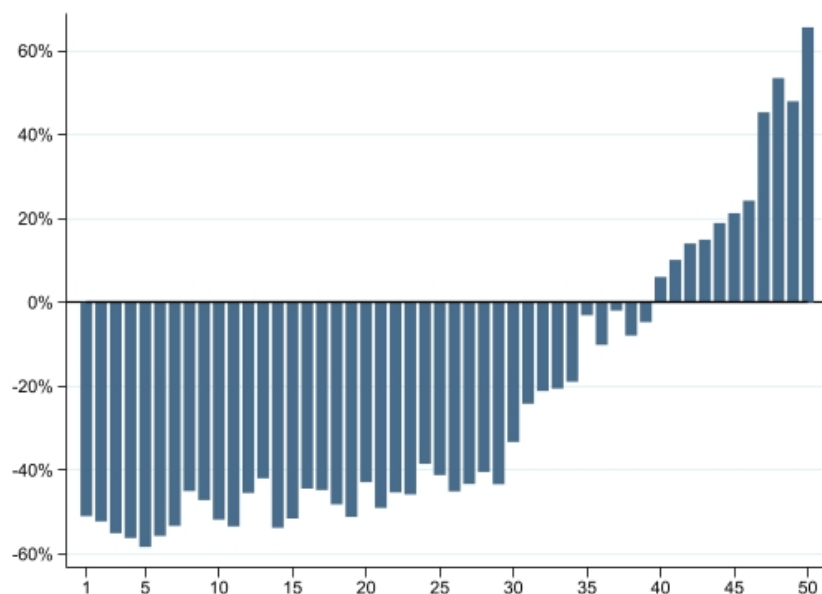
more in highly productive cities. This is because a larger wage gap increases the incentive to stay in or move to productive cities.

Table 9: Wage and Housing Price Dispersion

Characteristics	Baseline	Counterfactual	Difference (%)
Mean wage	1.18	1.44	0.27
Std. wage	0.12	0.25	0.13
Mean housing price	1.05	1.28	0.23
Std. housing price	0.49	0.65	0.16

Note: This table reports population-weighted mean and standard deviation of local wages and housing prices in the baseline and productivity-dispersion counterfactual economies. The last column reports the difference between the counterfactual and baseline economies.

Figure 8: Population Share Changes under the Counterfactual



Notes: This figure plots the percentage deviation in MSA-level population shares between the counterfactual and baseline economies. MSAs are ordered from low to high productivity levels.

Total Effect Table 10 reports migration values under the new environment. The overall distribution of migration values remains broadly similar to that in the baseline economy. Migration values continue to decline with age and increase with wealth. At the same time,

the relationship between migration values and tenure status remains similar to that in the baseline economy, with the exception of renters aged 50–64, who experience a substantially larger increase in migration value. Overall, the aggregate migration value rises from 2.56% in the baseline economy to 3.11% in the new environment.

However, the additional gains generated by the new environment are distributed unevenly across household groups. The largest increase occurs among households aged 50–64. For all households, the migration value rises by 1.15 points, compared with increases of only 0.04 points for households aged 20–34, 0.55 points for households aged 35–49, and 0.54 points for households aged 65 and above.

The additional gains also tend to be larger for wealthier households. Although the relationship is not monotonic within every age group, the aggregate results display a clear wealth gradient. The increase in migration value rises from 0.21 points for Q1 households to 0.92 points for Q4 households. Similarly, homeowners obtain larger gains than renters on average, with migration values increasing by 0.73 points for owners and 0.17 points for renters. Overall, the additional value generated by the new environment is concentrated among households aged 50–64, wealthier households, and homeowners.

Table 11 reports migration rates under the new environment. In contrast to migration values, migration rates generally decline. The overall migration rate falls from 2.55% in the baseline economy to 2.19% in the new environment. This decline reflects two changes generated by the counterfactual. First, the larger housing price gap across locations increases the cost of moving to productive cities. Second, the increase in productivity dispersion leads households to concentrate more heavily in highly productive cities, reducing the need for subsequent relocation.²⁶ As a result, migration becomes less frequent on average.

However, the decline in migration rates is not uniform across household groups. Migration rates become progressively higher relative to the baseline economy as age and wealth increase. For example, the aggregate migration rate falls by 1.21 points for households aged 20–34, but the decline becomes much smaller for households aged 35–49 and 50–64. Among households aged 65 and above, migration rates increase by 0.32 points relative to the baseline economy.

²⁶Recall that young and low-wealth households are the group most likely to move and they move to productive regions in the baseline economy. However, under the counterfactual, they are more likely to already be located in productive regions.

Table 10: Migration Value under New Environment

	Age Group				
	20–34	35–49	50–64	65+	Aggregate
Panel A. Migration Value					
<i>Wealth quartile by age</i>					
Q1	3.76%	2.53%	2.56%	0.81%	2.90%
Q2	3.80%	2.90%	2.83%	1.16%	2.62%
Q3	4.50%	3.47%	2.96%	1.64%	2.77%
Q4	6.08%	5.05%	4.08%	2.66%	3.88%
<i>Tenure status by age</i>					
Renter	3.91%	2.76%	3.36%	1.14%	3.21%
Owner	3.91%	3.76%	3.54%	1.88%	3.07%
Aggregate	3.91%	3.54%	3.53%	1.79%	3.11%
Panel B. Value Change (p.p.) from Baseline					
<i>Wealth quartile by age</i>					
Q1	0.05	0.28	0.96	0.24	0.21
Q2	-0.08	0.33	1.02	0.31	0.36
Q3	-0.06	0.39	0.95	0.43	0.55
Q4	0.12	0.73	1.16	0.69	0.92
<i>Tenure status by age</i>					
Renter	0.04	0.28	0.99	0.31	0.17
Owner	0.05	0.61	1.16	0.56	0.73
Aggregate	0.04	0.55	1.15	0.54	0.55

Notes: Panel A reports migration values measured by Consumption Equivalent Variation (CEV) under the productivity-dispersion counterfactual with pension benefits recalibrated to maintain the same replacement rate as in the baseline economy. Panel B reports percentage-point differences relative to the baseline economy.

A similar pattern appears across wealth groups. The aggregate migration rate declines by 0.79 points for Q1 households, compared with only 0.17 points for Q3 households, while the migration rate of Q4 households remains essentially unchanged. Thus, the reduction in migration is concentrated among younger and lower-wealth households, whereas migration rates are largely preserved among older and wealthier households.

Comparing Tables 10 and 11 reveals an important difference between migration values and migration rates. While migration rates decline overall, migration values increase for almost

all household groups. In particular, households aged 35–49 experience a substantial increase in migration values despite a noticeable decline in migration rates. These results suggest that migration rates alone do not provide a sufficient measure of the welfare gains generated by migration. Although households migrate less frequently in the new environment, the value of having the migration option increases because larger wage and housing price differences create more valuable relocation opportunities.²⁷

Intuition Why do some household groups obtain substantially larger additional gains from the new environment than others? I explain the results by considering how the effects vary across wealth and age groups. Since tenure status is closely related to wealth, I discuss the tenure results later.

Among low-wealth households, young households continue to migrate primarily for investment reasons.²⁸ A larger wage gap increases the gains from moving to productive locations. However, the larger housing price gap offsets these benefits. Table 12 illustrates this mechanism. Relative to the baseline economy, young low-wealth households experience larger gains in non-durable consumption but larger declines in housing size. This pattern suggests that higher labor income is used to compensate for the higher housing expenditures associated with expensive locations. As a result, the additional migration gains generated by the new environment remain relatively small.

The mechanism is different for middle-aged low-wealth households. These households migrate less for investment motives and more for reducing living costs. Therefore, they tend to move toward cheaper housing markets and use the lower housing prices to increase housing consumption. Since the housing price gap becomes larger in the counterfactual economy, the increase in housing size is larger than in the baseline economy. This housing size gain more than offsets the decline in non-durable consumption caused by moving to lower-wage locations. As a result, their migration gains become larger under the new environment. In

²⁷Since the counterfactual changes both wages and housing prices, it is also important to examine how migration values and migration rates respond to each of these changes separately. To do so, I consider counterfactuals in which only wages change and only housing prices change. The results are reported in the Appendix.

²⁸I report additional statistics related to migration motives, including wage and housing price changes at destinations, in the Appendix.

Table 11: Migration Rate under New Environment

	Age Group				
	20–34	35–49	50–64	65+	Aggregate
Panel A. Migration Rate					
<i>Wealth quartile by age</i>					
Q1	4.20%	1.75%	1.43%	1.23%	3.05%
Q2	2.77%	1.67%	1.60%	1.38%	1.91%
Q3	2.58%	1.70%	1.71%	1.51%	1.73%
Q4	2.70%	2.10%	1.98%	1.65%	1.92%
<i>Tenure status by age</i>					
Renter	4.18%	2.21%	1.99%	1.54%	3.41%
Owner	2.11%	1.69%	1.81%	1.49%	1.69%
Aggregate	3.67%	1.81%	1.82%	1.49%	2.19%
Panel B. Rate Change from Baseline					
<i>Wealth quartile by age</i>					
Q1	-1.11	-0.96	-0.37	0.25	-0.79
Q2	-1.32	-0.78	-0.14	0.31	-0.47
Q3	-1.19	-0.69	-0.05	0.34	-0.17
Q4	-1.04	-0.54	0.07	0.31	0.01
<i>Tenure status by age</i>					
Renter	-1.17	-0.98	-0.41	0.37	-0.89
Owner	-0.91	-0.61	0.02	0.32	-0.10
Aggregate	-1.21	-0.73	-0.01	0.32	-0.36

Notes: Panel A reports migration rates under the productivity-dispersion counterfactual with pension benefits recalibrated to maintain the same replacement rate as in the baseline economy. Panel B reports percentage-point differences relative to the baseline economy.

addition, these households are close to retirement, which makes wage losses from migration less important. After retirement, pension benefits do not vary across locations. Therefore, even when moving to a much cheaper housing market requires accepting much lower wages than the baseline, these households are more willing to make such moves if they can obtain a large increase in housing size.

The same logic helps explain the differences across wealth groups. Young high-wealth households also migrate to increase housing consumption. However, unlike older households, they

cannot completely ignore wage differences across locations. Despite their higher wealth, these households still have a long remaining working life and therefore continue to value future labor income.²⁹ As a result, they are reluctant to move to locations with very low housing prices if doing so requires giving up a substantial amount of future labor income. Consequently, they tend to relocate to moderately priced housing markets and obtain only moderate increases in housing size relative to older households.

For middle-aged high-wealth households, this constraint becomes much weaker. As above, future wage losses become less important for these households, while the benefits from larger housing consumption remain substantial. These households can therefore take greater advantage of the larger housing price gap by relocating to cheaper housing markets. Since housing consumption represents an important source of utility for wealthy households,³⁰ the larger housing size increase than the baseline economy brings large additional migration gains. In addition, wealthier households obtain larger additional migration gains than poorer households within the same age group because they can increase housing size by more after relocation. The same logic can be applied to retired households.

The tenure results broadly follow the wealth patterns. In particular, the sharp increase in migration values among renters aged 50–64 reflects their ability to relocate to substantially cheaper housing markets and obtain large housing size gains while placing less weight on future wage losses.

These mechanisms also help explain the migration-rate results. Under the new environment, young low-wealth households are more likely to already reside in productive cities, while moving into productive cities becomes more difficult because of the larger housing price gap. As a result, migration rates decline substantially for these households. In contrast, as households become older, larger wage losses associated with moving to cheaper locations become less important because they are closer to retirement. At the same time, wealthier households are better able to use the larger housing price gap to increase housing size. These two forces raise the net value of migration for older and wealthier households and increase

²⁹The Q4 group contains households with a wide range of wealth levels because wealth quartiles are constructed based on the population distribution. Consequently, not all households in the highest wealth quartile are extremely wealthy, and many still consider labor income an important determinant of consumption.

³⁰Recall that for wealthier households, a discrete increase in housing size can generate larger utility gains than additional non-durable consumption, which is already relatively high.

their incentive to relocate. Therefore, as households become older and wealthier, the decline in migration rates becomes smaller, and some groups even have higher migration rates than in the baseline economy

Table 12: Changes in Mover Gains under New Environment

	Age Group			
	20–34	35–49	50–64	65+
Panel A. Consumption Change from Baseline				
<i>Wealth quartile by age</i>				
Q1	7.61	3.36	-2.53	0.32
Q2	0.54	0.75	-2.77	0.28
Q3	-0.99	-1.38	-1.76	1.11
Q4	-3.37	-3.13	-1.59	1.28
<i>Tenure status by age</i>				
Renter	6.78	3.69	-3.02	0.07
Owner	-1.23	-2.49	-1.98	0.71
Panel B. Housing Size Change from Baseline				
<i>Wealth quartile by age</i>				
Q1	-6.13	-4.72	9.62	19.28
Q2	-0.53	-1.11	16.34	16.80
Q3	2.50	3.21	17.28	20.81
Q4	5.57	7.62	17.49	22.32
<i>Tenure status by age</i>				
Renter	-4.85	-3.35	13.70	24.42
Owner	-1.00	6.50	19.01	20.76

Notes: Panel A reports changes in movers' non-durable consumption relative to the baseline economy. Panel B reports changes in movers' housing size relative to the baseline economy. Values are reported in percentage points. Positive values indicate larger gains than in the baseline economy, while negative values indicate smaller gains. In the Appendix, I report changes in wages, housing prices at destinations, non-durable consumption, and housing size under the counterfactual economy for the reference.

6 Concluding Remarks

This paper develops a life-cycle quantitative spatial equilibrium model with household heterogeneity and endogenous location choice to study how the value of migration opportunities

varies across household groups. The results show that the distribution of migration values does not necessarily coincide with the distribution of migration rates. Low-wealth households migrate more frequently, but obtain relatively small welfare gains from migration, whereas high-wealth households migrate less often but receive substantially larger migration values. These findings suggest that the households who move most frequently are not necessarily the households who benefit most from migration opportunities.

This paper also examines how migration values change when regional wage and housing price dispersion increases. The results show that a larger regional gap raises migration values on average, but the additional gains are not distributed equally across households. In particular, aged households, high-wealth households, and homeowners obtain larger additional gains. In contrast, the relative distribution of migration values remains similar to that in the baseline economy, implying that the overall pattern of which household groups obtain relatively larger migration values changes little following the environmental change.

These findings have several policy implications. First, evaluating migration-related policies solely through changes in migration rates may provide an incomplete picture of their welfare consequences. Even when households face the same change in the economic environment, the welfare value of the migration option can differ substantially across household groups, and the value of migration opportunities cannot be fully inferred from observed migration behavior alone. Second, regional housing price dispersion plays an important role in determining the value of migration. The results show that a substantial portion of the value of migration arises through housing consumption adjustments, suggesting that changes in housing market conditions can have important implications for the welfare value of migration opportunities. Finally, because the additional migration values generated by greater regional dispersion are distributed unevenly across households, policy evaluations should consider not only average effects but also the distributional consequences across age, wealth, and housing tenure groups.

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